

No for Some, Yes for Others: Persona Prompts and Other Sources of False Refusal in Language Models

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Abstract

Large language models (LLMs) are increasingly integrated into our daily lives and personalized. However, LLM personalization might also increase unintended side effects. Recent work suggests that persona prompting can lead models to falsely refuse user requests. However, no work has fully quantified the extent of this issue. To address this gap, we measure the impact of 15 sociodemographic personas (based on gender, race, religion, and disability) on false refusal. To control for other factors, we also test 16 different models, 3 tasks (Natural Language Inference, politeness, and offensiveness classification), and nine prompt paraphrases. We propose a Monte Carlo-based method to quantify this issue in a sample-efficient manner. Our results show that as models become more capable, personas impact the refusal rate less and less. Certain sociodemographic personas increase false refusal in some models, which suggests underlying biases in the alignment strategies or safety mechanisms. However, we find that the model choice and task significantly influence false refusals, especially in sensitive content tasks. Our findings suggest that persona effects have been overestimated, and might be due to other factors.

loves to be around others.” to induce an extroverted persona (Jiang et al., 2023).

However, persona prompting can have unintended side effects on model behavior. Notably, previous works have shown that persona prompting can lead models to falsely refuse user requests based on sociodemographics or cultural factors (Gupta et al., 2024b; Plaza-del-Arco et al., 2024; de Araujo and Roth, 2024). False refusal, more generally, means models refuse safe requests, often because they superficially resemble unsafe prompts or mention sensitive topics (Röttger et al., 2024b; Chehbouni et al., 2024; Wang et al., 2024b). The disparity of false refusals across different sociodemographic personas creates unfair differences in user experiences and consequently reveals models’ underlying social biases.

To mitigate this problem, we first need to quantify it. This paper presents a large-scale study measuring the impact of prompting with different sociodemographic personas on false refusals. We include a total of 15 sociodemographic personas based on sociodemographic factors (gender, race, religion, and disability). To control for other contextual factors, we include a wide range of elements: three NLP tasks, 16 models, and nine prompt paraphrases. The models vary in size from small to medium and belong to different families, including Meta’s Llama (AI@Meta, 2024), Google’s Gemma (Team et al., 2024a) and Alibaba’s Qwen (Bai et al., 2023). The three tasks are 1) Natural Language Inference (NLI), where personas should not matter (so we expect no refusal), to increasing tasks that present sensitive content and thus are likely to produce refusal, namely 2) politeness and 3) offensiveness classification. The resulting combinatorial search space is massive and cannot be exhaustively mapped. We, therefore, propose a Monte Carlo-based method for measuring the impact of personas across model families on false refusals in a sample-efficient manner.

1 Introduction

Large language models (LLMs) are increasingly integrated into real-world applications, allowing users to interact with them in diverse ways, from creative writing to tutoring assistants. One way to improve user experience is through personalization, so that interactions are adapted to a user’s personal preferences, communication styles, and contextual needs (Rafieian and Yoganarasimhan, 2023; Salemi et al., 2024; Zhang et al., 2024). Recent works have shown the ability of LLMs to embody diverse personas in their responses through prompts like “You are a very friendly and outgoing person who

We find that personas and prompt variations matter more in early versions of the models. As they become more capable, these choices matter less. Instead, the choice of task and model has an increasing impact on the refusal results: some tasks and some model families trigger more refusals when prompted with specific personas (like Black, Muslim, and transgender), indicating potential biases within the models. Our findings suggest underlying biases in the alignment strategies and highlight the need for fairer alignment techniques that balance fairness and safety.

However, open-ended prompts elicit more refusals across tasks. Our results also show how often overlooked experimental design choices substantially influence model behavior, highlighting the need for more transparent reporting of researcher choices to improve reproducibility. Otherwise, we risk incorrectly ascribing causal effects to results that were influenced by researcher choices beyond what was studied. For example, prior studies on the impact of sociodemographic personas might have produced vastly different findings had they chosen a different task or studied different models.

Contributions: (i) We systematically evaluate the influence of sociodemographic persona variations on model refusal rates, controlling for task choice, prompt design, and model choice; (ii) We introduce a Monte Carlo sampling method to quantify the impact of different sources of refusals on model false refusal behavior. This allows us to efficiently measure how different sources shape false refusals in models. (iii) We quantify the impact of the various factors on false refusals through regression and Wasserstein-distance-based methods.

2 Sources of False Refusals

Our central research question is “*How much do persona choice and other experimental factors influence false refusal?*” Our starting hypothesis, based on prior work, is that *personas increase false refusals at least some of the time* (Gupta et al., 2024b; Plaza-del-Arco et al., 2024; de Araujo and Roth, 2024). However, *we do not expect all false refusals to be explained by personas*. Therefore, in addition to specific personas (§2.1), we control for other plausible sources of false refusal – specifically task choice (§2.2), prompt choice (§2.3), and model choice (§2.4).

2.1 Personas

Inspired by Gupta et al. (2024b), we consider 15 personas across four sociodemographic attributes: gender, race, religion, and disability. See Table 2 in Appendix A.1 for the full list of personas categorized by sociodemographics.

2.2 Tasks

We strongly suspect that the specific task influences refusal independent of persona: Tasks presenting logical content should not be affected. E.g., textual entailment should not depend on whether it was prompted by a Black woman or an Asian man. Meanwhile, more tasks that involve sensitive content might interact with personas. E.g., offensive language classification might very well depend on who is asking.

We choose three different classification tasks: **natural language inference** (NLI), which focuses on logical content, and two tasks involving sensitive content, which are **politeness classification** and **offensive language detection**. For NLI, the goal is to predict textual entailment, determining whether sentence A entails, contradicts, or is neutral with respect to sentence B. For this task, we select the **XNLI** dataset (Conneau et al., 2018) which is a multilingual version of the MultiNLI dataset (Williams et al., 2018) translated into 14 different languages. The dataset contains instances labeled as *entailment*, *contradiction*, and *neutral*.

In politeness classification, the task is to evaluate the politeness level of a given text on a scale from 0 to 5. Offensive language detection consists of rating how offensive a text is, also using a scale from 0 to 5. For both tasks, we use the **POPQUORN** (Potato-Prolific) dataset (Pei and Jurgens, 2023), which is a large-scale English dataset designed for several text-based tasks, including offensiveness and politeness rating. The offensiveness subset includes 13,036 annotated instances labeled on a scale from 1 (less offensive) to 5 (more offensive), while the politeness subset contains 25,042 annotated instances labeled on a scale from 1 (less polite) to 5 (more polite)¹.

2.3 Prompt Paraphrases

LLMs are known to be sensitive to the exact prompt phrasing and requested output format (Sclar et al., 2023; Scherrer et al., 2023; Röttger et al., 2024a).

¹Note: Task language is another plausible source of variance in model behavior. We focus on English-language tasks for feasibility reasons.

We introduce a total of nine prompt variations to explore how prompt design affects false refusals and its robustness to minimal changes. These variations focus on two key elements: phrasing and response format. For phrasing, we test three different ways of framing a question: “Given a text, classify it as...”, “Label this text as...”, and “Classify the following text as...”. For response format, we explore three types inspired by Röttger et al. (2024a): *unforced*, where the model can generate a detailed explanation, *semi-forced* where the model has to respond strictly with a label (e.g., “only answer with the label”) and *forced* where it must also choose a single option from a set (e.g., “you must pick one of the two options”).

Additionally, we have two further prompt setups: *persona* and *persona-free*. For the *persona*, the complete prompt comprises the persona description followed by the classification task. Tables 3 and 4 in Appendix A.2 show the list of prompt paraphrases. In contrast, the *persona-free* setup omits the persona description and directly presents the classification task.

2.4 Models

We test 16 open-weight LLMs across 9 popular model families, including state-of-the-art models as well as their prior iterations. This allows us to test how false refusal behaviors have evolved over time, as well as variance across model families and model scale. Specifically, we test the smallest and medium-sized versions of **Meta’s Llama** (AI@Meta, 2024), **Google’s Gemma** (Team et al., 2024a) and **Alibaba’s Qwen** (Bai et al., 2023). From the Llama family, we test six models from four generations: Llama2 in its 7B, and 13B versions (Touvron et al., 2023), Llama3-8B, Llama3.1-8B (AI@Meta, 2024), and Llama3.2 in its 1B and 3B versions (Meta, 2024). From the Qwen family, we include five models from three generations: Qwen1.5- $\{7B, 32B\}$, Qwen2-7B, and Qwen2.5- $\{7B, 32B\}$ (Wang et al., 2024a). From the Gemma family, we test five models from two generations: gemma- $\{2B, 7B\}$ (Team et al., 2024a), gemma-2- $\{2B, 9B, 27B\}$ (Team et al., 2024b). We evaluate the instruction-tuned versions of these models.

3 Experimental Setup

3.1 Monte Carlo Sampling Approach

When quantifying the impact of multiple experimental controls (e.g., prompt template and persona)

on model behavior (e.g., refusal rate), the amount of possible input combinations grows combinatorially with the number of experimental controls. In our setting, naively evaluating every possible combination of a prompt template $v \in V$ and persona $p \in P$ would result in a multiplicative factor of $|V| \times |P|$ per every input. Hence, conducting such controlled evaluations tends to be infeasible for a large number of experimental controls. Therefore, we introduce a nested Monte Carlo Sampling approach that allows us to explore in a sample-efficient manner how different experimental controls impact a model’s refusal behavior.

Let D represent the dataset containing texts $\{x_1, x_2, \dots, x_N\}$, where each x_n is associated with a label y_i for a specific task. Further, let P be the set of single-attribute sociodemographic personas $\{p_1, p_2, \dots, p_M\}$. The attributes span over four different classes (i.e., gender, race, religion and disability). Lastly, we have a set of prompt templates $T = \{t_1, t_2, \dots, t_K\}$ where each template t_k takes a persona p_m and a text x_n as an input and maps it to a final model input $s_{n,m,k} = t_k(x_n, p_m)$. By sampling at random a multiple tuples of (t_k, p_m) for every input text x_n , we can ensure a balanced distributions over prompt templates T and personas P in the final set of model inputs $\mathcal{S} = \{s_{n,m,k}\}$.

3.2 Sampling and Inference Setup

For each task, we initially select 1,000 samples per class (e.g., entailment, contradiction, and neutral for NLI). Each selected sample is then expanded into 10 variations by applying different personas (§2.1) and prompt paraphrases (§2.3). We ensure a balanced representation across these factors as discussed in §3.1. This results in a final dataset of 30,000 instances for NLI, 50,000 for politeness classification, and 47,010 for offensiveness classification. Tables 5 and 6 in Appendix B show the final data distribution of personas and prompt paraphrases for each task. We then prompt each of the 16 models on these datasets using task-specific prompt templates (§2.3). Each input prompt explicitly incorporates both the assigned persona and prompt variation relevant to the task. To ensure deterministic responses, we set the decoding temperature to 0 (greedy decoding).

3.3 Evaluation Setup

Quantifying refusals Models might refuse to comply with persona-assigned prompts for different reasons. For instance, (1) the inability to com-